

Branchseed Hackathon Playbook

A short, plain-language plan for maximizing score and delivering a memorable demo.

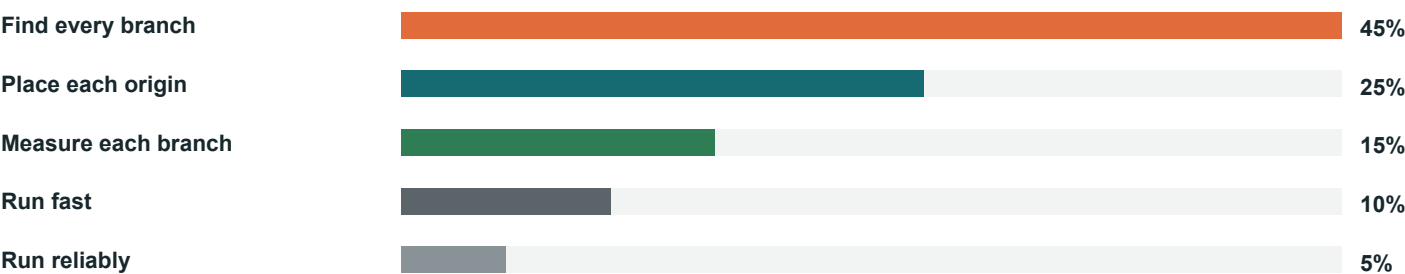
Build a branch finder, not a full vessel model.

The mission

The input is a contrast CT scan and a mask showing the main abdominal aorta. We must find every artery that starts directly from that aorta, then report each opening and its first few millimetres accurately.

Important correction: this is CT/CTA, not MRI. Do not spend time naming arteries or rebuilding the entire vessel tree; both are outside the core task.

Where the score comes from



What this means

- 70% of the score is finding the right branches and placing their origins correctly.
- A fast, reliable CPU solution matters, but accuracy comes first.
- The visual should help judges trust the detections; it must not slow down the scored program.

Terms in plain English

CT / CTA	A 3D X-ray scan. CTA uses contrast so blood inside arteries looks brighter.
NiftI	The file format that stores the 3D scan and its real-world position.
Aorta mask	A binary outline supplied by the organisers: 1 means main aorta, 0 means everything else.
Daughter artery	An artery that starts directly from the aorta.
Ostium	The centre of the opening where a daughter artery leaves the aorta.
Seed	A point in the middle of the daughter artery, exactly 5 mm from the ostium along its path.
Direction	A unit arrow showing which way the daughter artery initially travels.
Radius	The estimated distance from the seed to the inner vessel wall, in millimetres.
ROI / crop	A small area around the aorta. Processing only this area saves time and memory.
Tracking	Following the bright centre of a candidate artery for 5-10 mm.
Deduplication	Removing repeated detections so each real artery appears once.
Confidence	A simple score describing how strong and consistent the evidence is.

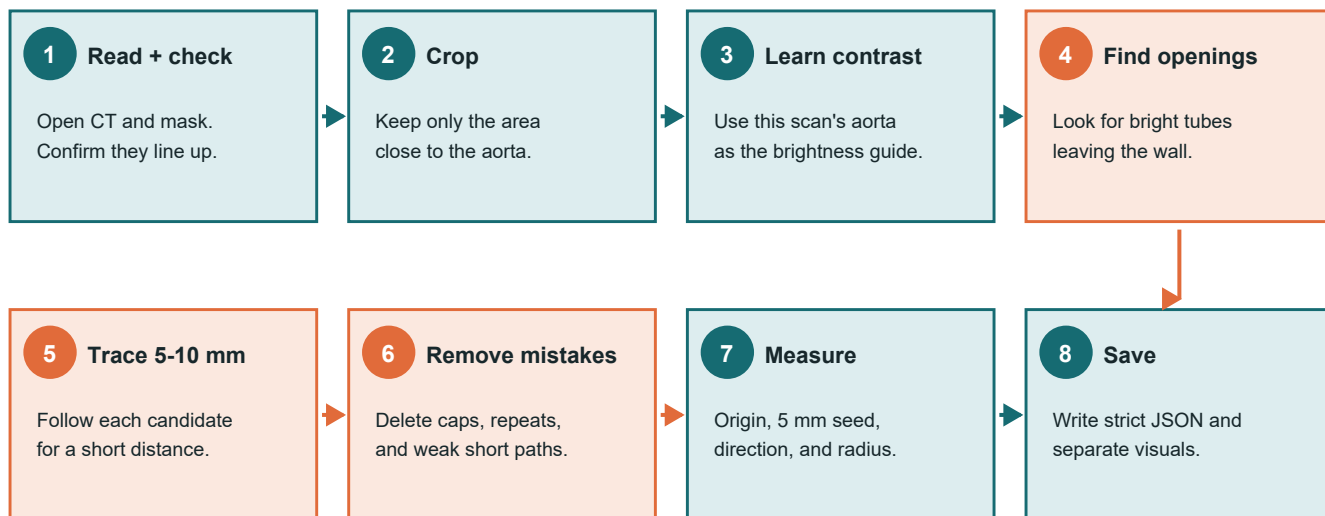
The required answer

For every case, save strict JSON containing one entry per branch: ostium coordinates, 5 mm seed coordinates, radius, and direction. All coordinates must be in real physical millimetres, not voxel numbers.

The best plan

The whole system, in one picture

Orange stages win most of the score: accurate discovery, short-path proof, and no duplicates.



Why this plan has the best chance

- It uses the supplied aorta mask as a powerful search guide instead of searching the whole body.
- It learns brightness from each scan, which handles the large contrast differences in the 25 cases.
- It proves that a candidate continues for at least 5 mm, which removes many false positives.
- It treats duplicates, cropped aorta ends, close openings, and common trunks as explicit rules.
- It runs on a laptop because every expensive step happens in a small crop.

The accuracy upgrade

Only after this pipeline works: if organisers provide reference outputs, train a very small classifier to rank proposed openings. Keep the geometry and tracking code deterministic. Do not train a large 3D model from scratch.

What to do next

36-hour build sequence

Always keep the last working version. New features must earn their place with a clear accuracy or demo gain.

0-3h Correct I/O	3-8h Fast baseline	8-16h Find openings	16-24h Track + measure	24-30h Visual + freeze	30-36h Offline test + demo
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Phase	Build now	Done when
1. Protect the submission	Robust `.nii` / `.nii.gz` loader, physical coordinates, exact CLI, strict JSON	A synthetic case produces correct, repeatable output
2. Create the fallback	Adaptive brightness threshold in a 25 mm shell, cap removal, connected components	All 25 cases run without crashing
3. Improve discovery	Surface normals, outward tube score, high-recall candidate list	Visible branches appear as candidates in both resolution groups
4. Prove each branch	Short beam tracker, ≥ 5 mm rule, first-split stop, radius plane	Every accepted branch has clear CT evidence
5. Remove repeats	Graph rules for same opening/path, nearby openings, and common trunks	One real branch produces one result
6. Freeze and demonstrate	Worst-case timing, memory test, offline install, three visual checks, Aorta Map	Clean 4-core/8 GB run and rehearsed five-minute demo

Four-person split

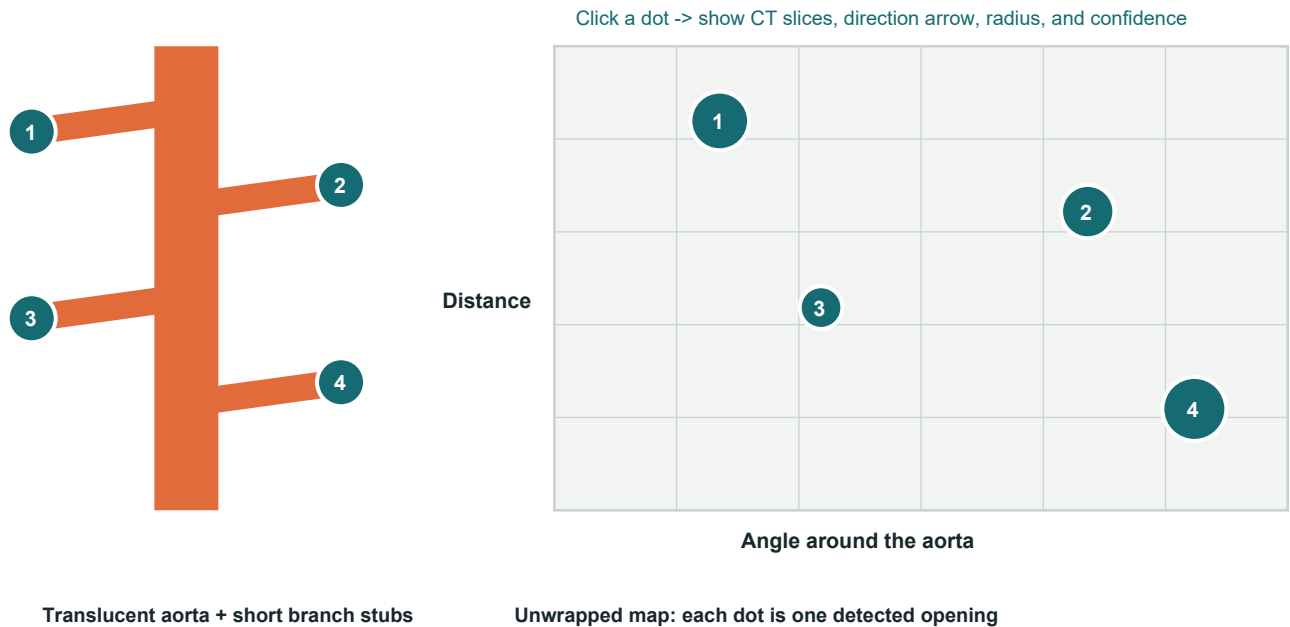
Person	Owns
A	File loading, coordinates, JSON, tests
B	Branch proposals and tracking
C	Manual review sheet, accuracy checks, runtime/memory
D	Aorta Map, branch evidence cards, demo

Rule: keep a frozen working version. Drop any feature that does not clearly improve branch accuracy, runtime, reliability, or demo clarity.

The visual that makes the project memorable

Creative visual: the Aorta Map

Turn a hard 3D structure into a simple map of where each branch starts.



Five-minute demo

Time	Show
0:00-0:35	The real task and score: direct branch discovery on CT, not full segmentation.
0:35-1:20	The eight-stage pipeline and why it fits CPU limits.
1:20-2:45	A difficult low-contrast or 1.5 mm case in the Aorta Map. Click one branch and show its CT evidence.
2:45-3:40	Close openings/common trunk handling and duplicate removal.
3:40-4:25	Runtime, memory, strict JSON, and a clean offline run.
4:25-5:00	One known failure, the fallback, and the final value: fast, measurable, explainable branch mapping.

Final win checklist

- Find as many real direct branches as possible without duplicates.
- Place origins in physical millimetres and make the 5 mm seed exact.
- Finish well under the runtime and memory limits on the hardest cases.
- Keep scored JSON simple; generate the Aorta Map separately.
- Do not build vessel naming, full-tree reconstruction, disease diagnosis, CFD, or a large neural model.

Source basis

Condensed from the full Branchseed strategy report and the official six-page Toralis Labs challenge document. Key technical references: SimpleITK physical transforms; ITK vesselness and fast marching; published CTA ostium localization; VMTK/SlicerVMTK centreline and cross-section tools.